Computer Networking: Formulas and Solved Numericals

Chapters 1, 2 and 3 (Kurose & Ross, Top-Down Approach slides)

Format of every problem: Question, Short explanation, Solution. Chapter 3 is limited to the requested topics: transport-layer services, multiplexing/demultiplexing, UDP and its header and checksum, rdt 1.0, 2.0, 2.1, 2.2, 3.0, rdt 3.0 utilisation and rdt 3.0 with pipelining. Where a slide gives a setup without numbers, the problem is solved symbolically and says so.

PART A: FORMULAS (CHAPTER WISE)

Chapter 1: Introduction

- Transmission delay: $d_{trans} = L / R$ (L = packet length in bits, R = link rate in bps)
- Propagation delay: $d_{prop} = d / s$ (d = link length, s = propagation speed, about 2×10^8 m/s)
- Nodal delay: $d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$
- Traffic intensity = $L \cdot a / R$ (a = average packet arrival rate in packets/sec). Near 0: small queueing delay; near 1: large delay; above 1: delay grows without bound
- Circuit switching: number of users = link rate / rate per user
- Packet switching, N users each active with probability p: $P(n \text{ active}) = C(N, n)\, p^n\, (1 - p)^{(N-n)}$
- End-to-end throughput on one path = $\min(R_c, R_s)$
- Per-connection throughput with 10 connections sharing a backbone link $R = \min(R_c, R_s, R/10)$

Chapter 2: Application Layer

- Non-persistent HTTP, per object: response time = 2 RTT + file transmission time
- Persistent HTTP: one TCP setup, then 1 RTT per object; with pipelining about 1 RTT for all referenced objects

Chapter 3: Transport Layer (requested topics)

- Internet checksum: add all 16-bit words, wrap any carry-out back into the sum (wraparound), then take the one's complement of the result
- Receiver check for UDP: recompute the checksum; if it differs from the checksum field, an error is detected
- Stop-and-wait (rdt 3.0) transmission time: $D_{trans} = L / R$
- Stop-and-wait utilisation: $U_{sender} = (L/R) / (RTT + L/R)$
- Effective throughput (stop-and-wait) = $L / (RTT + L/R)$
- Pipelined utilisation with N packets in flight: $U_{sender} = N \cdot (L/R) / (RTT + L/R)$
- Packets in flight needed for full utilisation (U = 1): $N = (RTT + L/R) / (L/R)$

Multiplexing/demultiplexing, the UDP header, rdt 1.0, 2.0, 2.1 and 2.2 are protocol descriptions with no numerical problems on the slides, so they add no solved problems below.

PART B: SOLUTIONS

Chapter 1

Q1.1: One-hop transmission delay

Question: A packet has L = 10 Kbits and the link rate is R = 100 Mbps. Find the one-hop transmission delay.

Short explanation: Transmission delay is the time needed to push all bits of the packet onto the link.

Solution:

- $L = 10 \times 10^3 = 10{,}000$ bits
- $R = 100 \times 10^6 = 10^8$ bps

$$d_{trans} = L / R = 10^4 / 10^8 = 10^{-4} \text{ s}$$

Answer: 0.1 msec

Q1.2: Caravan analogy, toll booths 100 km apart

Question: A ten-car caravan (a 10-bit packet) passes toll booths 100 km apart. A booth takes 12 sec per car and cars travel at 100 km/hr. How long until the caravan is lined up before the 2nd booth?

Short explanation: This is transmission delay plus propagation delay. The booth is the link transmitter and the highway is the propagation path.

Solution:

- Transmission time = 12 × 10 = 120 sec = 2 min
- Propagation time = 100 km / (100 km/hr) = 1 hr = 60 min

$$\text{Total} = 120 \text{ sec} + 1 \text{ hr} = 2 \text{ min} + 60 \text{ min}$$

Answer: 62 minutes

Q1.3: Caravan analogy with faster cars

Question: Cars now travel at 1000 km/hr and a booth takes 1 min per car. Will cars arrive at the 2nd booth before all cars are serviced at the 1st booth?

Short explanation: Compare the arrival time of the first car at booth 2 with the time booth 1 needs to service all ten cars.

Solution:

- First car is serviced at 1 min
- Propagation = 100 km / (1000 km/hr) = 0.1 hr = 6 min
- First car reaches booth 2 at 1 + 6 = 7 min
- By 7 min booth 1 has serviced 7 cars, so 10 − 7 = 3 cars are still at booth 1

Answer: Yes. The first car arrives at booth 2 while three cars are still at booth 1.

Q1.4: Packet switching versus circuit switching

Question: A 1 Gb/s link is shared by users who each need 100 Mb/s when active and are active 10% of the time. How many users can be supported under circuit switching and under packet switching?

Short explanation: Circuit switching reserves capacity for every user. Packet switching shares capacity, so it supports more users because they are rarely all active together.

Solution:

- Circuit switching: 1000 Mb/s / 100 Mb/s = 10 users
- Packet switching with N = 35, p = 0.1: the link overloads only if more than 10 users are active at once

$$P(\text{more than 10 active}) = \Sigma_{n=11}^{35} C(35, n)\,(0.1)^n\,(0.9)^{35-n} \approx 0.00042$$

- This is about 0.04%. The slide quotes it as "less than .0004"; the exact value is 0.00042, so the slide's figure is rounded.

Answer: circuit switching 10 users; packet switching 35 users (more than 3 times as many) with a very small chance of overload.

Q1.5: Throughput

Question: A server sends a file through a path with server link R_s and client link R_c. What is the average end-to-end throughput? With 10 connections sharing a backbone link R, what is the per-connection throughput?

Short explanation: The slowest link on the path limits throughput. This is the bottleneck link.

Solution:

- If $R_s < R_c$, throughput = R_s. If $R_s > R_c$, throughput = R_c.

$$\text{Throughput} = \min(R_s, R_c)$$

$$\text{With 10 shared connections: } \min(R_c, R_s, R/10)$$

Answer: the bottleneck link sets the throughput.

Chapter 2

Q2.1: Non-persistent HTTP response time (single object)

Question: Find the HTTP response time for one object using non-persistent HTTP.

Short explanation: Non-persistent HTTP opens a new TCP connection for every object.

Solution:

- 1 RTT to set up the TCP connection
- 1 RTT for the HTTP request and the first bytes of the response
- Plus the file transmission time

Answer: response time = 2 RTT + file transmission time

Q2.2: Web page with a base HTML file and 10 JPEG images

Question: A page contains text and references to 10 JPEG images. Find the total response time for the different HTTP modes. The slide gives no numbers, so the answers are in terms of RTT and transmission times (T).

Short explanation: Count how many RTTs each mode needs for the base file plus 10 objects.

Solution:

- Non-persistent, sequential: 11 objects × 2 RTT = 22 RTT + sum of all T

- Non-persistent, 10 parallel connections: 2 RTT (base) + 2 RTT (all images in parallel) = 4 RTT + transmission times
- Persistent without pipelining: 2 RTT (base, includes setup) + 10 × 1 RTT = 12 RTT + transmission times
- Persistent with pipelining: 2 RTT (base) + about 1 RTT (all 10 requests together) = about 3 RTT + transmission times

Answer: 22 RTT, 4 RTT, 12 RTT and about 3 RTT respectively (plus transmission times).

Chapter 3 (requested topics only)

Q3.1: UDP / Internet checksum

Question: Compute the checksum of two 16-bit words: 1110 0110 0110 0110 and 1101 0101 0101 0101.

Short explanation: The sender adds the words, wraps any carry-out around into the sum, and takes the one's complement. The result goes in the UDP checksum field.

Solution:

Step 1: add the two words.

```
  1110 0110 0110 0110
+ 1101 0101 0101 0101
---------------------
1 1011 1011 1011 1011   (17 bits: carry-out of 1)
```

Step 2: wraparound, add the carry-out back to the 16-bit result.

```
  1011 1011 1011 1011
+                   1
---------------------
  1011 1011 1011 1100   (sum)
```

Step 3: one's complement, flip every bit of the sum.

```
  1011 1011 1011 1100
  0100 0100 0100 0011   (checksum)
```

Check in hexadecimal: E666 + D555 = 1BBBB, wraparound gives BBBB + 1 = BBBC, complement = 4443, which equals 0100 0100 0100 0011.

Answer: sum = 1011101110111100, checksum = 0100010001000011

Q3.2: Receiver check and why the checksum is weak protection

Question: (a) How does the receiver verify the segment from Q3.1? (b) Flip bits in the two words so that the checksum does not change.

Short explanation: The receiver adds all words including the checksum. An all-ones result means no error was detected. Errors that cancel out in the sum go unnoticed.

Solution:

(a) Receiver verification: sum + checksum should be all ones.

```
  1011 1011 1011 1100   (sum of data words)
+ 0100 0100 0100 0011   (checksum)
---------------------
  1111 1111 1111 1111   (no error detected)
```

(b) Flip the last two bits of each word, as shown on the slide.

- First word: 1110 0110 0110 0110 becomes 1110 0110 0110 0101 (value decreases by 1)
- Second word: 1101 0101 0101 0101 becomes 1101 0101 0101 0110 (value increases by 1)

Sum = 1110 0110 0110 0101 + 1101 0101 0101 0110 = 1 1011 1011 1011 1011

After wraparound the sum is again 1011 1011 1011 1100, so the checksum is again 0100 0100 0100 0011.

Answer: the two bit flips cancel, the checksum is unchanged, and the receiver cannot detect the error. This is why the checksum is called weak protection.

Q3.3: rdt 3.0 (stop-and-wait) utilisation

Question: A 1 Gbps link has 15 msec propagation delay and 8000-bit packets (RTT = 30 msec). Find the transmission time, the sender utilisation and the effective throughput.

Short explanation: The sender transmits one packet, then waits a full RTT for the ACK before sending the next one, so the link is idle most of the time.

Solution:

Step 1: transmission time.

$$D_{trans} = L / R = 8000 \text{ bits} / 10^9 \text{ bps} = 8 \times 10^{-6} \text{ s} = 8 \text{ µs} = 0.008 \text{ msec}$$

Step 2: utilisation.

$$U_{sender} = (L/R) / (RTT + L/R) = 0.008 / (30 + 0.008) = 0.008 / 30.008 \approx 0.00027$$

Step 3: effective throughput, one packet per 30.008 msec.

$$\text{Throughput} = 8000 \text{ bits} / 0.030008 \text{ s} \approx 266{,}600 \text{ bps} \approx 33 \text{ kB/sec}$$

Answer: D_{trans} = 8 µs, U_{sender} ≈ 0.00027 (0.027%), throughput ≈ 33 kB/sec on a 1 Gbps link. The protocol limits the use of the physical link.

Q3.4: rdt 3.0 with pipelining

Question: With the same link as Q3.3, find the utilisation if the sender pipelines 3 packets. Then find how many packets must be in flight to fully use the link (the second part is derived from the formula, not given on the slide).

Short explanation: Pipelining lets the sender transmit several packets before the first ACK returns, so the link is busy for a larger fraction of each RTT.

Solution:

Part 1: three packets in flight.

$$U_{sender} = 3 \cdot (L/R) / (RTT + L/R) = 3 \times 0.008 / 30.008 = 0.024 / 30.008 \approx 0.00080$$

This is exactly 3 times the stop-and-wait value of 0.00027.

Part 2: full utilisation (U = 1).

$$N \cdot (L/R) = RTT + L/R$$

$$N = (RTT + L/R) / (L/R) = 30.008 / 0.008 = 3751$$

Answer: U ≈ 0.00080 with 3 packets (a factor of 3 improvement); about 3751 packets in flight give full utilisation.